SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Online Pet Shop Management System

Requirements Baseline – Version 1.0

Document Information	Details
Project	Online Pet Shop Management System
Organization	_____
Document	Software Requirements Specification
Version	1.0
Baseline	Requirements Baseline v1.0
Development Approach	Incremental Development with Controlled Change Management
Technology	HTML, CSS, JavaScript, PHP, MySQL
Development Environment	XAMPP
Status	Proposed for Review
Prepared By	A. Paheershana (JF/IOCT/24/04)
Date	_____

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the requirements for Version 1.0 of the Online Pet Shop Management System. The document establishes the agreed requirements that will be used as the reference for system analysis, system design, database design, software development, testing, and release activities.

The requirements contained in this document represent the initial proposed scope of the system. Once reviewed and approved by the relevant stakeholders, this document will become the Requirements Baseline v1.0. Any new requirement introduced after the Version 1.0 baseline has been approved shall be handled through the Software Change Management process using a formal Change Request (CR).

1.2 Project Background

The proposed Online Pet Shop Management System is a web-based application intended to manage pet shop products and customer orders online. Customers will be able to register, log in, browse pet products, view products by category, add products to a shopping cart, and place orders. The administrator will manage users, categories, products, stock, and customer orders through the administration module.

The system is intended to provide a simple and organized method for managing pet shop activities and to make it more convenient for customers to browse and purchase pet products online.

2. Problem Statement

- Pet product information may be maintained using manual or disconnected records.
- Managing customer details, product information, stock, and orders manually can be time-consuming.
- Manual stock management may result in inaccurate available quantities.
- Customers may need to visit the physical shop to check product availability and prices.
- Manual order processing can lead to errors in order and stock records.
- There is a need for a centralized web-based system for managing pet shop information and customer orders.

The proposed system aims to address these problems by providing a centralized platform for product, stock, customer, cart, and order management.

3. Project Objectives

- To develop a web-based system for managing an online pet shop.
- To provide customers with an easy way to browse pet products.
- To organize pet products according to different categories.
- To allow customers to add products to a shopping cart and place orders.
- To manage product information, prices, and stock efficiently.
- To automatically reduce product stock when a customer successfully purchases a product.
- To prevent customers from ordering quantities greater than the available stock.
- To manage customer orders in an organized manner.
- To provide an administration module for managing the system.
- To reduce manual work in managing pet shop records.

4. Project Scope

4.1 In Scope for Version 1.0

Authentication

- Admin login and logout.
- Customer registration, login, and logout.
- Credential validation.

Administration

- Admin dashboard.
- Manage users.
- Manage categories.
- Manage products.
- Manage product prices and stock.
- Manage customer orders.
- Update order status.

User Management

- Register customer.
- View customer profile.
- Update profile information.
- View order history.

Category Management

- Add, view, update, and delete categories.
- Display products according to category.

Product Management

- Add, view, update, and delete products.
- Manage product prices.
- Manage product stock.
- Upload product images.
- Check available stock.

Shopping Cart & Order Management

- Add products to cart.
- View and update cart quantities.
- Remove products from cart.
- Calculate total price.
- Validate stock before order placement.
- Place orders.
- Automatically reduce stock after a successful order.
- View order details and order status.

Technical Scope

- HTML, CSS, and JavaScript frontend.
- PHP backend.

- MySQL database.
- XAMPP development environment.
- Responsive web interface.

4.2 Out of Scope for Version 1.0

- Online payment gateway integration.
- Delivery tracking and live courier integration.
- Advanced sales analytics and forecasting.
- Discount/coupon management.
- Product reviews and ratings.
- Email/SMS notification integration.
- Multi-vendor marketplace functionality.
- Mobile application.
- Advanced role-based access control.
- Production-scale deployment.

These features may be considered for future releases through approved Change Requests.

5. Stakeholders

5.1 Stakeholder Register

Stakeholder	Role	Interest / Responsibility
Administrator	System User	Manage users, categories, products, stock, and orders
Customer	End User	Browse products, manage cart, and place orders
Pet Shop Owner / Management	Business Stakeholder	Monitor shop operations, products, stock, and orders
Project Team	Development Team	Analyze, design, develop, test, and maintain the system
Lecturer / Supervisor	Academic Supervisor	Review project progress, requirements, and software engineering practices

6. User Roles

6.1 Administrator

The administrator is responsible for managing the main shop information and operations. The administrator shall be able to:

- Log in and log out.
- Manage users.
- Add, view, update, and delete categories.
- Add, view, update, and delete products.
- Manage product prices and stock.
- View customer orders.
- Update order status.

6.2 Customer

The customer uses the system to browse and purchase pet products. The customer shall be able to:

- Register and log in.
- View and update their profile.
- Browse products and categories.
- View product details and available stock.
- Add products to the shopping cart.
- Update or remove cart items.
- Place orders when sufficient stock is available.
- View order details and order history.
- Log out.

Customers shall not be permitted to manage product, category, stock, or other customers' records.

7. Functional Requirements

Functional requirements describe what the system shall do.

7.1 Authentication Requirements

FR-01 – User Login

The system shall allow an authorized administrator or registered customer to log in using valid credentials.

FR-02 – Login Validation

The system shall validate the credentials provided by the user.

FR-03 – Invalid Login Handling

The system shall display an appropriate error message when invalid login credentials are provided.

FR-04 – User Logout

The system shall allow an authenticated user to log out.

FR-05 – Customer Registration

The system shall allow a new customer to create an account using required information.

7.2 Administration Requirements

FR-06 – Admin Dashboard

The system shall provide an administrator dashboard for accessing management functions.

FR-07 – Manage Users

The system shall allow the administrator to view and manage registered customer accounts.

FR-08 – Manage Categories

The system shall allow the administrator to add, view, update, and delete product categories.

FR-09 – Manage Products

The system shall allow the administrator to add, view, update, and delete pet product records.

FR-10 – Manage Product Price

The system shall allow the administrator to add and update product prices.

FR-11 – Manage Product Stock

The system shall allow the administrator to add and update the available stock quantity of products.

FR-12 – Product Image Upload

The system shall allow the administrator to upload product images.

FR-13 – Manage Orders

The system shall allow the administrator to view customer orders and update their order status.

7.3 Customer & Product Requirements

FR-14 – View Products

The system shall allow customers to browse available pet products.

FR-15 – View by Category

The system shall allow customers to view products according to their selected category.

FR-16 – View Product Details

The system shall display relevant product information including name, price, description, image, and available stock.

FR-17 – View Profile

The system shall allow an authenticated customer to view their own profile.

FR-18 – Update Profile

The system shall allow an authenticated customer to update permitted profile information.

7.4 Shopping Cart & Order Requirements

FR-19 – Add to Cart

The system shall allow customers to add an available product to the shopping cart.

FR-20 – Update Cart Quantity

The system shall allow customers to change the quantity of an item in the shopping cart, subject to available stock.

FR-21 – Remove Cart Item

The system shall allow customers to remove a product from the shopping cart.

FR-22 – Calculate Total

The system shall calculate the total price based on product prices and selected quantities.

FR-23 – Check Stock Availability

The system shall verify that sufficient stock is available before accepting an order.

FR-24 – Place Order

The system shall allow a customer to place an order when all required information is valid and sufficient stock is available.

FR-25 – Automatic Stock Reduction

The system shall automatically reduce the available stock quantity by the purchased quantity after a successful order is placed.

FR-26 – Prevent Overselling

The system shall not allow a customer to place an order for a quantity greater than the available stock.

FR-27 – Order Details

The system shall allow customers to view their order details and order status.

FR-28 – Order History

The system shall allow authenticated customers to view their previous orders.

8. Non-Functional Requirements

8.1 Security

NFR-01 – Password Security

The system shall store passwords using secure password hashing rather than plain text.

NFR-02 – Authentication

Protected system functions shall require user authentication.

NFR-03 – Authorization

The system shall restrict administrative functions to authorized administrators.

NFR-04 – Data Validation

The system shall validate required user, product, cart, and order input before processing.

NFR-05 – Response Time

Under normal operating conditions, commonly used pages and operations should respond within approximately three seconds.

NFR-06 – Database Performance

The database shall support efficient retrieval and modification of product, stock, user, and order records.

NFR-07 – User Interface

The system shall provide a simple, understandable, and consistent user interface.

NFR-08 – Validation Messages

The system shall provide clear messages when input is invalid, stock is insufficient, or an operation cannot be completed.

NFR-09 – Data Integrity

The system shall maintain consistency between order quantities and product stock.

NFR-10 – Error Handling

The system shall handle expected application errors without causing unexpected termination.

NFR-11 – Maintainability

The system shall use a modular structure so that future approved changes can be implemented with minimal unnecessary impact.

NFR-12 – Browser Compatibility

The web application should operate correctly on commonly used modern web browsers.

NFR-13 – Responsive Interface

The user interface should adapt to common desktop, tablet, and mobile screen sizes.

9. User Stories

US-01 – Login

As a registered user, I want to log in using my credentials, so that I can access the functions available to me.

US-02 – Register

As a customer, I want to create an account, so that I can purchase products and view my orders.

US-03 – Manage Products

As an administrator, I want to manage products, so that accurate product information is available to customers.

US-04 – Manage Stock

As an administrator, I want to manage product stock, so that available quantities remain accurate.

US-05 – Browse Products

As a customer, I want to browse pet products by category, so that I can find products easily.

US-06 – Add to Cart

As a customer, I want to add products to my cart, so that I can purchase selected products.

US-07 – Place Order

As a customer, I want to place an order, so that I can purchase products online.

US-08 – Automatic Stock Update

As a shop administrator, I want stock to automatically decrease after a successful purchase, so that stock records remain accurate.

US-09 – Prevent Overselling

As a customer, I want the system to check stock before ordering, so that I cannot order unavailable quantities.

US-10 – View Order History

As a customer, I want to view my order history, so that I can track my previous purchases.

US-11 – Manage Orders

As an administrator, I want to view and update customer orders, so that I can manage order processing.

US-12 – Logout

As an authenticated user, I want to log out, so that my account is protected when I finish using the system.

10. Acceptance Criteria

AC-01 – Login

Given a registered user is on the login page, when valid credentials are entered and Login is selected, then the system shall authenticate the user and display the appropriate page. Invalid credentials shall display an appropriate error message.

AC-02 – Add Product

Given the administrator is logged in, when valid product information is entered and saved, then the product shall be stored and displayed in the product list.

AC-03 – Update Stock

Given the administrator is logged in, when a valid stock quantity is entered, then the product stock shall be updated and the new quantity shall be stored in the database.

AC-04 – Add to Cart

Given a customer is viewing an available product, when the customer selects Add to Cart, then the selected product and quantity shall be added to the cart.

AC-05 – Insufficient Stock

Given a customer requests a quantity greater than available stock, when the customer attempts to order, then the system shall prevent the order and display an insufficient-stock message.

AC-06 – Place Order and Reduce Stock

Given a customer has a valid cart and sufficient stock is available, when the customer confirms the order, then the order shall be stored successfully and the product stock shall automatically decrease by the purchased quantity.

AC-07 – Order Details

Given a customer has placed an order, when the customer opens the order history, then the system shall display the order details and status.

AC-08 – Admin Order Update

Given the administrator is authenticated, when the administrator updates an order status, then the new status shall be stored and displayed.

AC-09 – Profile

Given a customer is authenticated, when the customer opens the profile page, then the system shall display that customer's own profile information.

AC-10 – Logout

Given a user is authenticated, when Logout is selected, then the authenticated session shall end and the user shall return to the login page.

11. MoSCoW Prioritization

Priority	Requirements
Must Have	Login, Logout, Customer Registration, Product CRUD, Category CRUD, Stock Management, Product Browsing, Shop
Should Have	Product Search, Improved Validation, User Feedback, Order History
Could Have	Basic Dashboard, Product Image Improvements
Won't Have in v1.0	Online Payment Gateway, Delivery Tracking, Reviews/Ratings, Discounts/Coupons, Advanced Analytics, Notifications

12. Product Backlog

ID	Product Backlog Item	Priority	Related Requirements	Initial Status
US-01	User Login	Must	FR-01, FR-02, FR-03	Planned
US-12	User Logout	Must	FR-04	Planned
US-02	Customer Registration	Must	FR-05	Planned
US-03	Manage Products	Must	FR-09, FR-10, FR-12	Planned
US-04	Manage Stock	Must	FR-11	Planned
US-05	Browse Products	Must	FR-14, FR-15, FR-16	Planned
US-06	Shopping Cart	Must	FR-19, FR-20, FR-21, FR-22	Planned
US-07	Place Order	Must	FR-23, FR-24	Planned
US-08	Automatic Stock Update	Must	FR-25	Planned
US-09	Prevent Overselling	Must	FR-26	Planned
US-10	Order History	Should	FR-27, FR-28	Planned
US-11	Manage Orders	Must	FR-13	Planned

13. Use Cases

13.1 Use Case List

Use Case ID	Use Case	Primary Actor
UC-01	Login	Administrator / Customer
UC-02	Register Customer	Customer
UC-03	Manage Categories	Administrator
UC-04	Manage Products	Administrator
UC-05	Manage Stock	Administrator
UC-06	Browse Products	Customer
UC-07	Manage Shopping Cart	Customer
UC-08	Place Order	Customer
UC-09	Automatically Reduce Stock	System
UC-10	View Order History	Customer
UC-11	Manage Orders	Administrator
UC-12	Logout	Administrator / Customer

13.2 Use Case Example – Place Order

Field	Description
Use Case ID	UC-08
Name	Place Order
Actor	Customer
Preconditions	Customer must be authenticated; cart must contain products; sufficient stock must be available.
Trigger	Customer selects Place Order / Confirm Order.
Main Flow	Review cart → Validate customer/order data → Check stock → Calculate total → Save order → Save order items → F
Alternative Flow	Insufficient stock, invalid data, or order processing error.
Postcondition	Order is stored successfully and the corresponding product stock is reduced.

14. Business Rules

The following business rules apply to Version 1.0.

- BR-01 – Only authenticated users may access protected system functionality.
- BR-02 – Only authorized administrators may add, update, or delete product and category records.
- BR-03 – Each product shall have a unique product identifier.
- BR-04 – Product price shall be a valid non-negative value.
- BR-05 – Product stock quantity shall be a non-negative whole number.
- BR-06 – Required product, customer, and order information must be provided before saving.
- BR-07 – Customers may purchase only quantities that are currently available in stock.
- BR-08 – The system shall verify stock availability before confirming an order.
- BR-09 – After a successful order, product stock shall be reduced by the purchased quantity.

- BR-10 – Stock shall not become negative as a result of an order.
- BR-11 – If an order cannot be successfully completed, the corresponding stock quantity shall not be reduced.
- BR-12 – Customers may view their own order history but shall not manage other customers' orders.
- BR-13 – Passwords must not be stored in plain text.

15. Data Requirements

Version 1.0 requires the system to maintain the following core information.

15.1 User Information

- User ID
- Name
- Email / Username
- Password
- Phone Number
- Address
- Role

15.2 Category Information

- Category ID
- Category Name
- Description

15.3 Product Information

- Product ID
- Product Name
- Category ID
- Price
- Stock Quantity
- Description
- Product Image

15.4 Order Information

- Order ID
- Customer ID
- Order Date
- Total Amount
- Order Status

15.5 Order Item Information

- Order Item ID
- Order ID
- Product ID

- Quantity
- Unit Price
- Subtotal

The detailed database structure, relationships, normalization, constraints, and SQL schema will be documented separately during the Database Design phase.

16. System Constraints

- The application will be developed as a web-based system.
- HTML, CSS, and JavaScript will be used for the frontend.
- PHP will be used for the backend.
- MySQL will be used as the database.
- XAMPP will be used as the development environment.
- Version 1.0 will focus on essential online pet shop management functionality.
- New requirements after baseline approval must follow the Change Request process.

17. Assumptions

- The pet shop administrator will provide valid product, category, price, and stock information.
- Users will have access to a modern web browser and internet/local network access as applicable.
- Customers will provide valid information during registration and ordering.
- Product stock information entered by the administrator will represent the actual available stock.
- The system will be used for a single pet shop in Version 1.0.
- Future functionality can be introduced through controlled software changes.
- Requirements may evolve after Version 1.0 and will therefore be managed through the Software Change Management process.

18. Requirements Traceability Matrix

The Requirements Traceability Matrix (RTM) provides traceability between functional requirements, user stories, acceptance criteria, and future testing activities.

Requirement	User Story	Acceptance Criteria	Test Case
FR-01 Login	US-01	AC-01	TC-01
FR-05 Registration	US-02	AC-01	TC-02
FR-09 Product Management	US-03	AC-02	TC-03
FR-11 Stock Management	US-04	AC-03	TC-04
FR-14/15/16 Browse Products	US-05	AC-04	TC-05
FR-19–22 Shopping Cart	US-06	AC-04	TC-06
FR-23/24 Place Order	US-07	AC-06	TC-07
FR-25 Automatic Stock Reduction	US-08	AC-06	TC-08
FR-26 Prevent Overselling	US-09	AC-05	TC-09
FR-27/28 Order History	US-10	AC-07	TC-10
FR-13 Manage Orders	US-11	AC-08	TC-11

Requirement	User Story	Acceptance Criteria	Test Case
FR-04 Logout	US-12	AC-10	TC-12

The test case identifiers will be formally defined and expanded during the Test Planning and Testing sessions.

19. Requirements Review

Before establishing the baseline, the requirements shall be reviewed with the relevant stakeholders.

Review Question 1

Can customers directly modify product stock?

Decision: No. Only authorized administrators can manage product stock.

Review Question 2

What happens when a customer orders more than the available stock?

Decision: The system shall prevent the order and display an insufficient-stock message.

Review Question 3

Should product stock automatically decrease after a successful purchase?

Decision: Yes. Stock shall be automatically reduced by the purchased quantity.

Review Question 4

Can product stock become negative?

Decision: No. The system shall prevent negative stock quantities.

Review Question 5

Is online payment included in Version 1.0?

Decision: No. Online payment integration is outside the Version 1.0 scope.

Review Question 6

Can customers manage other customers' orders?

Decision: No. Customers may view only their own order history.

Review Question 7

Are advanced reports included?

Decision: No. Advanced reporting is outside the Version 1.0 baseline.

20. Version 1.0 Requirements Baseline

After stakeholder review and approval, the requirements contained in this document shall become the Requirements Baseline v1.0. The baseline establishes the official reference point for Version 1.0 development.

- New requirements shall not be added directly to the Version 1.0 scope.
- Any new requirement must be submitted through the Software Change Management process.
- Approved changes shall be assessed for their impact on requirements, design, database, implementation, testing, and documentation.
- The relevant baselines and version information shall be updated when an approved change is implemented.

The general change process is: New Requirement → Change Request → Impact Analysis → Change Approval → Implementation → Testing → Code Review → Merge → Documentation Update → New Version / Release.

21. Configuration and Version Control

The approved requirements document shall be maintained under version control where applicable.

21.1 Baseline Information

Item	Value
Baseline Name	Requirements Baseline v1.0
Version	1.0
Status	Proposed for Review / Approved after Sign-off
Baseline Tag	requirements-v1.0
Baseline Date	_____
Approved By	_____

The baseline tag requirements-v1.0 will provide a fixed reference point for future change analysis.

22. Future Change Management

Once the Version 1.0 requirements have been baselined, future requirements may be introduced by the pet shop owner/client through the formal change management process.

Example Future Request

"The pet shop now requires an online payment gateway."

This requirement shall not be automatically added to Version 1.0. Instead, the request shall follow the formal change process:

Customer / Owner Request → CR-001 → Impact Analysis → Project Approval → Requirements Update → Design Update → Database Update → Backend Update → Frontend Update → Testing → Release v1.1

23. Version History

Version	Date	Description	Author	Status
0.1	_____	Initial requirements draft	A. Paheershana	Draft
0.2	_____	Requirements reviewed and revised	A. Paheershana	Review
1.0	_____	Requirements baseline established	A. Paheershana	Approved

24. Approval

By approving this document, the relevant stakeholders confirm that the requirements defined for Version 1.0 represent the agreed initial scope of the Online Pet Shop Management System.

Role	Name	Signature	Date
Client / Business Representative	_____	_____	_____
Project Team Representative	A. Paheershana	_____	_____
Lecturer / Supervisor	_____	_____	_____

Appendix A – Version 1.0 Scope Summary

Included

- Authentication
- Admin management
- Customer registration and profile
- Category management
- Product management
- Product price and stock management
- Automatic stock reduction after successful purchase
- Stock availability validation
- Shopping cart
- Order placement
- Order management
- Order history
- Basic security
- Responsive web interface
- PHP backend
- MySQL database

Not Included

- Online payment gateway
- Delivery tracking
- Product reviews and ratings
- Discount/coupon management
- Advanced analytics and reporting
- Notifications
- Mobile application
- Multi-vendor marketplace
- Advanced access control
- Production-scale deployment

Appendix B – Initial Technology Environment

Component	Technology
Frontend	HTML, CSS, JavaScript
Backend	PHP
Database	MySQL
Development Environment	XAMPP
Browser	Modern web browser

Appendix C – Project Lifecycle

The initial project shall follow the following lifecycle:

Project Scenario → Requirements Engineering → Requirements Baseline → System Analysis & Design → Design Baseline → Database Design → Database Baseline → Backend Development → Frontend Development → Integration → Testing → Version 1.0 Release

After the first release: Change Request → Impact Analysis → Approved Change → Implementation → Testing → New Release

Appendix D – Baseline Statement

Requirements Baseline v1.0

The requirements contained in this document constitute the proposed/approved Version 1.0 requirements baseline for the Online Pet Shop Management System. Any modification to these requirements after baseline approval must be controlled through the Software Change Management process.

Baseline Information

Baseline Tag	**requirements-v1.0**
Baseline Date	_____
Approved By	_____
Approval Signature	_____

Appendix E – Project Document Sequence

The Requirements Baseline is positioned within the overall practical project as follows:

01 – Initial Client Scenario → 02 – Initial Project Proposal → 03 – Project Plan / Practical Roadmap → 04 – SRS + Requirements Baseline v1.0 → Session 3 – SCM Setup → Session 4 – System Design → Session 5 – Database Design → Development → Testing → Version 1.0 Release → CR-001 → Impact Analysis → Version 1.1

Final Baseline Note

The Version 1.0 scope has intentionally been kept focused on the essential online pet shop functions: authentication, user management, category management, product and stock management, shopping cart, and order management. A key business rule is that product stock must be checked before an order is accepted and automatically reduced by the purchased quantity after a successful order. This requirement provides accurate stock information and helps prevent overselling.

Features such as online payment, delivery tracking, reviews, discounts, notifications, and advanced analytics are excluded from the Version 1.0 baseline and may be introduced through formal Change Requests in future releases.